The winding of DC machines shorts from the first coil side and finally ended with the same and hence the DC machines windings are generally called as "closed coil winding".

In making winding table for lap winding, it is important that we must add back pitch to obtain back end side and subtract front pitch to obtain front end side.

9.5.12 Double layer simplex wave winding diagram

Develop a double layer simplex wave winding diagram for a 2 poles, 6 slots, DC machine having 6 commutator segments. Indicate the position of brushes.

Solution

Number of poles P = 2

Number of slots S = 6

Number of commutator segments = 6

We know

No. of coils C = No. of commutator segments

Hence C = 6

$$\text{Back pitch } Y_b = \frac{2C}{P} \pm K$$

$$= \frac{2 \times 6}{2} \times K$$

$$= 6+1 = 7 \text{ (or)}$$

$$6-1 = 5$$

$$\text{Coil pitch } Y = \frac{2C \pm 2m}{P/2}$$

m = 1 (for simplex)

$$\text{Then } Y = \frac{2 \times 6 \pm 2 \times 1}{2/2}$$

$$= 12 \pm 2$$

$$= 12 + 2 = 14 \text{ (or)}$$

$$12 - 2 = 10$$

Front pitch $Y_f = Y - Y_b$

Also $Y_f = Y_b$

So $Y_f = 14 - 7 = 7$ (or)

$10 - 5 = 5$

Selected Y_b and Y_f are

$Y_b = Y_f = 7$ (or)

$(Y_f = Y_f = 5)$

Winding table ($Y_f = Y_f = 5$)

Sl. No.	$Y_b = 5$	$Y_f = 5$
1	1 – 6	6 – 11
2	11 – 4	4 – 9
3	9 – 2	2 – 7
4	7 – 12	12 – 5
5	5 – 10	10 – 3
6	3 – 8	8 – 1

In making winding table for wave winding, it is important that we must add back pitch and front pitch to obtain back end side and front end side respectively.

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